Chapter 4 – Transmission Lines

1 Chapter 4: Transmission Lines and Signal Integrity

1.1 4.1 Preview: Why Transmission Lines Matter *(chap4.pdf, p.3)*

At high speeds, an interconnect is no longer a simple wire --- it behaves as a transmission line because:

1. **Time delay:** a signal takes a finite, nonzero time $T_D = L/v$ to travel from one end to the other.
2. **Reflections:** any impedance mismatch causes reflected waves that distort the signal --- the dominant mechanism of **signal integrity** degradation. *(chap4.pdf, p.3)*

Signal integrity means the waveforms at input and output of the line are identical (or approximately so). *(chap4.pdf, p.3)*

1.1.1 Types of Transmission Lines *(chap4.pdf, pp.4--5)*

Structure	Dispersive?	Notes
Two-wire	Yes	Outer insulator causes dispersion
Wire above ground	Yes	Outer insulator causes dispersion
Coaxial cable	No	Fields entirely inside; no dispersion
Stripline (PCB inner layer)	No	Fields entirely inside dielectric
Microstrip (PCB outer layer)	Yes	Partial air, partial dielectric
Coplanar strip (no inner planes)	Yes	Partial air, partial dielectric

(chap4.pdf, pp.4--5)

1.2 4.2 Transmission-Line Equations (Lossless)

1.2.1 Distributed Model *(chap4.pdf, p.6)*

The lossless line is modeled as an infinite cascade of inductors ($l\,\Delta z$ H) in series and capacitors ($c\,\Delta z$ F) in shunt, where l and c are the **per-unit-length** inductance [H/m] and capacitance [F/m].

Physical meaning *(chap4.pdf, p.7)*: - l and c cause a finite transit time: $T_D = L/v$ - At GHz frequencies, conductor resistance r (due to skin effect) can become significant

1.2.2 First-Order Coupled Equations *(chap4.pdf, pp.8--9)*

Applying KVL around a Δz section:

$$-\frac{\partial V}{\partial z} = l\,\frac{\partial I}{\partial t}$$

(chap4.pdf, p.8)

Applying KCL at the shunt capacitor node:

$$-\frac{\partial I}{\partial z} = c\,\frac{\partial V}{\partial t}$$

(chap4.pdf, p.9)

These are the **coupled transmission-line equations** (first-order form). *(chap4.pdf, p.9)*

1.2.3 Second-Order Wave Equations *(chap4.pdf, p.10)*

Decoupling the two equations above:

$$\frac{\partial^2 V}{\partial z^2} = lc\frac{\partial^2 V}{\partial t^2}$$

(chap4.pdf, p.10)

$$\frac{\partial^2 I}{\partial z^2} = lc\frac{\partial^2 I}{\partial t^2}$$

(chap4.pdf, p.10)

These are called the **uncoupled transmission-line equations** (second-order form). *(chap4.pdf, p.10)*

1.3 4.3 Per-Unit-Length Parameters

1.3.1 TEM Mode Properties *(chap4.pdf, pp.11--12)*

Transmission lines support the **Transverse Electromagnetic (TEM) mode**: $H_z = 0$ and $E_z = 0$. This allows static (dc) field solutions to be used for l and c even for time-varying excitations. *(chap4.pdf, p.11)*

Faraday's law ($H_z = 0$):

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

(chap4.pdf, p.11)

Ampere's law ($E_z = 0$):

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

(chap4.pdf, p.11)

For a **homogeneous** medium surrounding the conductors, l and c satisfy: *(chap4.pdf, p.12)*

$$lc = \mu\varepsilon = \mu_0\varepsilon_0\mu_r\varepsilon_r$$

(chap4.pdf, p.12)

from which the phase velocity:

$$v = \frac{1}{\sqrt{lc}} = \frac{1}{\sqrt{\mu\varepsilon}} = \frac{v_0}{\sqrt{\mu_r\varepsilon_r}}$$

(chap4.pdf, p.12)

1.3.2 Effective Relative Permittivity *(chap4.pdf, p.13)*

For **inhomogeneous** media (e.g., microstrip with partial air and partial dielectric), an effective permittivity ε_r' is defined such that an equivalent homogeneous medium would give the same l and c:

$$lc = \mu_0\varepsilon_0\varepsilon_r'$$

(chap4.pdf, p.13)

This gives $v = v_0/\sqrt{\varepsilon_r'}$ and $Z_C = \sqrt{l/c} = (1/v)\sqrt{1/\varepsilon_0\mu_0} \cdot 1/\sqrt{\varepsilon_r'/\mu_r}$.

1.3.3 Wire-Type Structures

Magnetic Field around a Current-Carrying Wire ***(chap4.pdf, p.14)*** By Ampere's law (current uniformly distributed, cylindrical symmetry):

$$H_\phi = \frac{I}{2\pi r}$$

(chap4.pdf, p.14)

Magnetic Flux through a Surface *(chap4.pdf, p.15)* Per unit length of line, the flux threading between two radii r_1 and r_2:

$$\Psi = \frac{\mu_0 I}{2\pi} \ln\left(\frac{r_2}{r_1}\right)$$

(chap4.pdf, p.15)

Used to find l via $\Psi = l\,I$. *(chap4.pdf, p.15)*

Electric Field around a Charged Wire *(chap4.pdf, p.16)* By Gauss' law (charge q per unit length, cylindrical symmetry):

$$E_r = \frac{q}{2\pi\varepsilon r}$$

(chap4.pdf, p.16)

Voltage between Two Points *(chap4.pdf, p.17)*

$$V = \frac{q}{2\pi\varepsilon} \ln\left(\frac{r_2}{r_1}\right)$$

(chap4.pdf, p.17)

Used to find c via $q = c\,V$. *(chap4.pdf, p.17)*

Two-Wire Line *(chap4.pdf, pp.18--22)* To avoid the proximity effect, assume widely separated wires (wire radius a, separation d, with $a \ll d$).

Total magnetic flux between the two wires per unit length:

$$\Psi \approx \frac{\mu_0 I}{\pi} \ln\left(\frac{d}{a}\right)$$

(chap4.pdf, p.18)

Approximate per-unit-length inductance (valid for $d \gg a$):

$$l \approx \frac{\mu_0}{\pi} \ln\left(\frac{d}{a}\right) \quad [\mathrm{H/m}]$$

(chap4.pdf, p.19)

Exact per-unit-length inductance:

$$l = \frac{\mu_0}{\pi} \cosh^{-1}\left(\frac{d}{2a}\right) \quad [\mathrm{H/m}]$$

(chap4.pdf, p.20)

Exact per-unit-length capacitance:

$$c = \frac{\pi\varepsilon_0}{\cosh^{-1}(d/2a)} \quad [\mathrm{F/m}]$$

(chap4.pdf, p.20)

Voltage between the two wires (sum of contributions from each wire):

$$V = \frac{q}{\pi\varepsilon} \ln\left(\frac{d}{a}\right)$$

(chap4.pdf, p.21)

Approximate per-unit-length capacitance (same result as simplifying the exact formula):

$$c \approx \frac{\pi\varepsilon_0}{\ln(d/a)} \quad [\mathrm{F/m}]$$

(chap4.pdf, p.22)

Wire above a Ground Plane ***(chap4.pdf, pp.23--24)*** Using the **method of images**: the ground plane is replaced by an image wire at distance $2h$ below the original. The capacitance between the wire and ground plane equals **twice** the capacitance of two wires with separation $2h$: *(chap4.pdf, p.23)*

Exact per-unit-length capacitance (wire height h above ground, wire radius a):

$$c = \frac{2\pi\varepsilon_0}{\cosh^{-1}(h/a)} \quad [\mathrm{F/m}]$$

(chap4.pdf, p.24)

Exact per-unit-length inductance (half of two-wire result, by reciprocity):

$$l = \frac{\mu_0}{2\pi}\cosh^{-1}\left(\frac{h}{a}\right) \quad [\mathrm{H/m}]$$

(chap4.pdf, p.24)

Coaxial Cable ***(chap4.pdf, pp.25--27)*** No proximity effect due to symmetry (inner radius a, outer radius b).

From Ampere's law:

$$H_\phi = \frac{I}{2\pi r}, \quad a \le r \le b$$

(chap4.pdf, p.25)

Flux per unit length:

$$\Psi = \frac{\mu_0 I}{2\pi}\ln\left(\frac{b}{a}\right)$$

(chap4.pdf, p.25)

Exact per-unit-length inductance:

$$l = \frac{\mu_0}{2\pi}\ln\left(\frac{b}{a}\right) \quad [\mathrm{H/m}]$$

(chap4.pdf, p.26)

Exact per-unit-length capacitance (using $lc = \mu_0\varepsilon_0\varepsilon_r$):

$$c = \frac{2\pi\varepsilon_0\varepsilon_r}{\ln(b/a)} \quad [\mathrm{F/m}]$$

(chap4.pdf, p.26)

Alternatively, voltage between inner and outer conductors:

$$V = \frac{q}{2\pi\varepsilon}\ln\left(\frac{b}{a}\right)$$

(chap4.pdf, p.27)

giving the same $c = 2\pi\varepsilon/\ln(b/a)$. *(chap4.pdf, p.27)*

1.3.4 PCB Structures *(chap4.pdf, pp.28--32)*

In PCB work, the key parameters are the **characteristic impedance** and **velocity**: *(chap4.pdf, p.28)*

$$Z_C = \sqrt{\frac{l}{c}}, \qquad v = \frac{1}{\sqrt{lc}} = \frac{v_0}{\sqrt{\varepsilon_r'}}$$

(chap4.pdf, p.28)

PCB geometries do not have closed-form exact solutions; approximate formulas are used.

Stripline ($t = 0$) ***(chap4.pdf, p.29)*** The inner strip is fully embedded in a uniform dielectric ($\varepsilon_r' = \varepsilon_r$):

$$Z_C = \frac{60}{\sqrt{\varepsilon_r}} \ln\left(\frac{4b}{\pi d}\right) \quad [\Omega]$$

(chap4.pdf, p.29)
where b = board thickness between ground planes, d = strip width.

Microstrip ($t = 0$) ***(chap4.pdf, p.30)*** The outer strip is on the surface with air above and dielectric below:

$$Z_C = \frac{120\pi}{\sqrt{\varepsilon_r'}\,(W/h + 1.393 + 0.667\ln(W/h + 1.444))} \quad [\Omega]$$

(chap4.pdf, p.30)
Effective relative permittivity:

$$\varepsilon_r' = \frac{\varepsilon_r + 1}{2} + \frac{\varepsilon_r - 1}{2}\left(1 + \frac{12h}{W}\right)^{-1/2}$$

(chap4.pdf, p.30)
where W = strip width, h = substrate height.

Coplanar Strip ($t = 0$) ***(chap4.pdf, p.31)***

$$Z_C \approx \frac{120\pi}{\sqrt{\varepsilon_r'}} \cdot \frac{K(k')}{K(k)}$$

(chap4.pdf, p.31)
Effective permittivity:

$$\varepsilon_r' = 1 + \frac{\varepsilon_r - 1}{2} \cdot \frac{K(k')K(k_1)}{K(k)K(k_1')}$$

(chap4.pdf, p.31)

Strips on Opposite Sides of the Board ($t = 0$) ***(chap4.pdf, p.32)*** Wide lands on opposite sides give **lower characteristic impedance** than same-side placement, because the facing area creates more capacitance. This is exploited for power planes, which need low impedance ($V = L\,di/dt$, so low L minimizes noise). *(chap4.pdf, p.32)*

1.4 4.4 Time-Domain Solution

1.4.1 General Solution for the Lossless Line *(chap4.pdf, p.34)*

The general solution consists of forward- ($+z$) and backward-($-z$) traveling waves:

$$V(z,t) = V^+(t - z/v) + V^-(t + z/v)$$

(chap4.pdf, p.34)

$$I(z,t) = \frac{V^+(t - z/v) - V^-(t + z/v)}{Z_C}$$

(chap4.pdf, p.34)
where:

$$Z_C = \sqrt{\frac{l}{c}}, \qquad v = \frac{1}{\sqrt{lc}}$$

(chap4.pdf, p.34)

1.4.2 Reflection Coefficients *(chap4.pdf, pp.35--37)*

Load reflection coefficient (at $z = L$):

$$\Gamma_L = \frac{Z_L - Z_C}{Z_L + Z_C}$$

(chap4.pdf, p.35)

Source reflection coefficient (at $z = 0$):

$$\Gamma_S = \frac{Z_S - Z_C}{Z_S + Z_C}$$

(chap4.pdf, p.36)

The **current reflection coefficient is the negative** of the voltage reflection coefficient. *(chap4.pdf, p.35)*

Initial forward-traveling voltage launched from the source at $t = 0$:

$$V_{\text{init}}^{+} = V_S \frac{Z_C}{Z_S + Z_C}$$

(chap4.pdf, p.37)

where V_S is the source open-circuit voltage. The process of repeated reflections continues: a wave reflecting at the load re-reflects at the source, and so on. *(chap4.pdf, p.37)*

1.4.3 Bounce (Lattice) Diagram *(chap4.pdf, pp.42--43)*

The lattice diagram is a graphical tool with: - Horizontal axis: position z (0 to L) - Vertical axis: time t

Each line segment represents a traveling wave. The total voltage or current at any (z, t) is the superposition of all waves that have reached that point. *(chap4.pdf, p.42)*

When the line is matched at both ends ($\Gamma_S = \Gamma_L = 0$), all reflections vanish and only a time delay $T_D = L/v$ is imposed --- the line "doesn't matter." *(chap4.pdf, p.43)*

1.5 4.5 High-Speed Digital Interconnects and Signal Integrity

1.5.1 Clock Skew *(chap4.pdf, p.44)*

Unequal time delays for signals reaching different logic gates cause **clock skew**. At high speeds, these delays are on the order of rise/fall times and become critical in the system timing budget.

1.5.2 Impedance Mismatch Effects *(chap4.pdf, pp.45--46)*

Any change in line cross-section (and thus Z_C) causes reflections. Common sources: - Different trace widths - Vias (signal transitions between PCB layers) - Connector transitions

Result: ringing --- the load voltage oscillates around the final value, potentially causing false logic transitions. *(chap4.pdf, p.46)*

Typical CMOS gate parameters: output resistance 10--30 Ω, input capacitance 5--15 pF. *(chap4.pdf, p.46)*

1.5.3 Load Voltage with Mismatch *(chap4.pdf, p.48)*

For a line connecting two gates ($Z_S \neq Z_C$, $Z_L \neq Z_C$):

$$V_L(t) = V_{\text{init}}^{+}(1 + \Gamma_L) \sum_{k=0}^{n} (\Gamma_S \Gamma_L)^k \cdot u(t - (2k+1)T_D)$$

(chap4.pdf, p.48)

The sign of the reflection coefficients determines whether the signal overshoots or undershoots: - Positive Γ_L: may cause **multiple activations** of the gate *(chap4.pdf, p.49)* - Negative Γ_L: may **postpone activation** *(chap4.pdf, p.49)*

1.5.4 Capacitive Termination *(chap4.pdf, pp.50--52)*

For a source-matched line ($Z_S = Z_C$) terminated with a capacitor C_L:

The load voltage in the Laplace domain:

$$\hat{V}_L(s) = \frac{V_0}{s} \frac{1}{1 + sZ_C C_L}$$

(chap4.pdf, p.51)

Inverse transform (time domain):

$$V_L(t) = V_0 \left(1 - e^{-t/\tau_C}\right) u(t - T_D)$$

(chap4.pdf, p.52)

with time constant:

$$\tau_C = Z_C\, C_L$$

(chap4.pdf, p.50)

Behavior: initially the capacitor looks like a **short circuit**, gradually transitioning to an **open circuit**. The capacitive load introduces additional time delay (measured at the 50% point) $t_d = 0.693\,\tau_C$. *(chap4.pdf, p.52)*

This is also the basis of **Time Domain Reflectometry (TDR)**: by observing the reflected waveform, one can characterize the load. *(chap4.pdf, p.50)*

1.5.5 Inductive Termination *(chap4.pdf, p.53)*

For a source-matched line terminated with an inductance L_L, the time constant is:

$$\tau_L = \frac{L_L}{Z_C}$$

(chap4.pdf, p.53)

Behavior: initially the inductor looks like an **open circuit**, gradually transitioning to a **short circuit**. *(chap4.pdf, p.53)*

1.5.6 Matching Schemes *(chap4.pdf, pp.54--63)*

Series Match ***(chap4.pdf, p.54)*** A resistor R is inserted in series at the source such that:

$$R_S + R = Z_C$$

(chap4.pdf, p.54)

Results: 1. Load voltage equals source level (full signal swing) *(chap4.pdf, p.54)* 2. The resistor dissipates **no power** in steady state (no dc current through the line) *(chap4.pdf, p.54)*

With a capacitive load, the total delay at the 50% point consists of: *(chap4.pdf, p.57)* 1. Source rise time / 2 2. One-way line delay T_D 3. Capacitive load delay $0.693\,Z_C C_L$

Parallel Match ***(chap4.pdf, pp.55--58)*** A resistor R in parallel with the load such that $R \parallel Z_L = Z_C$ (approximately, $R = Z_C$ for high-impedance load). *(chap4.pdf, p.55)*

Results: 1. Load voltage is **less than** source level *(chap4.pdf, p.55)* 2. Resistor **consumes power** even in steady state *(chap4.pdf, p.55)*

With a capacitive load: the parallel match fails at high frequencies because the capacitor dominates and acts as a short circuit. Logic errors may result. *(chap4.pdf, p.58)*

1.5.7 When Does the Line Not Matter? *(chap4.pdf, pp.59--63)*

Criterion 1: Electrical Length ***(chap4.pdf, p.59)***

$$L < \frac{\lambda}{10} \quad \text{at the highest significant frequency}$$

(chap4.pdf, p.59)

Criterion 2: Rise Time vs. Time Delay ***(chap4.pdf, p.59)***

$$\tau_r > 10\,T_D$$

(chap4.pdf, p.59)

If the pulse rise time is greater than 10 times the one-way line delay, line effects do not significantly degrade the output waveform. *(chap4.pdf, p.59)*

Simulation results *(chap4.pdf, pp.60--63)*: | τ_r/T_D | Overshoot | |----------|----------| | 1 | 7 V (for 5 V supply) | | 5 | 6 V | | 10 | 5.3 V | | 20 | 5.2 V |

As τ_r increases: maximum frequency decreases ($f_{max} \propto 1/\tau_r$), high-frequency content is reduced, and ringing diminishes. *(chap4.pdf, p.63)*

1.5.8 Discontinuities *(chap4.pdf, pp.64--72)*

Different Characteristic Impedances ***(chap4.pdf, pp.64--65)*** At a junction between line 1 (Z_{C1}) and line 2 (Z_{C2}):

Reflection coefficient (incident from left):

$$\Gamma_{12} = \frac{Z_{C2} - Z_{C1}}{Z_{C2} + Z_{C1}}$$

(chap4.pdf, p.64)

Transmission coefficient (voltage transmitted into line 2):

$$T_{12} = 1 + \Gamma_{12} = \frac{2Z_{C2}}{Z_{C1} + Z_{C2}}$$

(chap4.pdf, p.65)

Similarly from right to left:

$$\Gamma_{21} = \frac{Z_{C1} - Z_{C2}}{Z_{C1} + Z_{C2}} = -\Gamma_{12}, \qquad T_{21} = 1 + \Gamma_{21}$$

(chap4.pdf, p.65)

Eliminating Reflections at Discontinuities ***(chap4.pdf, pp.66--67)***

- **Series match at source only:** does not eliminate reflections at an internal junction. *(chap4.pdf, p.66)*
- **Parallel match at load only:** does not eliminate reflections either. *(chap4.pdf, p.67)*
- **Series match at source + parallel match at load:** eliminates all reflections. *(chap4.pdf, p.67)*

Series Multiple Feedlines (Fan-Out) ***(chap4.pdf, pp.68--69)*** A series match at the source ($R_S + R = Z_C$) eliminates all reflections if the feedline characteristic impedances are equal. *(chap4.pdf, p.68)*

A parallel match at a midpoint junction does **not** eliminate reflections --- it creates a new impedance mismatch: *(chap4.pdf, p.69)*

$$Z_{\text{junction}} = Z_{C2} \parallel Z_{C3} \neq Z_{C1}$$

(chap4.pdf, p.69)

Parallel Multiple Feedlines ***(chap4.pdf, pp.70--72)*** Initial forward voltage for two parallel lines from one source:

$$V_{\text{init}}^{+} = V_S \frac{Z_{C1} \parallel Z_{C2}}{Z_S + Z_{C1} \parallel Z_{C2}}$$

(chap4.pdf, p.70)

Reflection and transmission at the parallel junction *(chap4.pdf, p.71)*:

When $T_{D1} = T_{D2}$ and $R_S = Z_C/2$: the reflected waves from the two lines cancel each other ($\Gamma_{12} = -T_{12} = -T_{21}$). *(chap4.pdf, p.71)*

Best matching scheme for parallel lines: **parallel match at each load** (place Z_C across each load). This is less power-efficient than series match. *(chap4.pdf, p.72)*

1.6 4.6 Sinusoidal (Phasor) Analysis of the Line

1.6.1 Lossless Line --- Phasor Form *(chap4.pdf, pp.73--78)*

From the time-domain solution, substituting sinusoidal steady-state $e^{j\omega t}$:

$$\hat{V}(z) = V^{+}\, e^{-j\beta z} + V^{-}\, e^{+j\beta z}$$

(chap4.pdf, p.73)

$$\hat{I}(z) = \frac{V^{+}\, e^{-j\beta z} - V^{-}\, e^{+j\beta z}}{Z_C}$$

(chap4.pdf, p.73)

where:

$$Z_C = \sqrt{\frac{l}{c}}, \qquad \beta = \omega\sqrt{lc}$$

(chap4.pdf, p.74)

Voltage reflection coefficient at position z:

$$\Gamma(z) = \frac{V^{-}}{V^{+}}\, e^{+2j\beta z}$$

(chap4.pdf, p.74)

At $z = L$ (load): $\Gamma_L = (Z_L - Z_C)/(Z_L + Z_C)$.

Input impedance at distance d from the load:

$$Z_{in}(-d) = Z_C \frac{Z_L + jZ_C \tan(\beta d)}{Z_C + jZ_L \tan(\beta d)}$$

(chap4.pdf, p.75)

Or equivalently in terms of Γ:

$$Z_{in}(z) = Z_C \frac{1 + \Gamma(z)}{1 - \Gamma(z)}$$

(chap4.pdf, p.75)

VSWR (voltage standing-wave ratio) --- a measure of reflection:

$$S = \frac{1 + |\Gamma_L|}{1 - |\Gamma_L|}$$

(chap4.pdf, p.77)

When matched ($\Gamma_L = 0$): $S = 1$ (no reflections). *(chap4.pdf, p.77)*

Average power flowing in the $+z$ direction:

$$P_{av} = \frac{|V^{+}|^2}{2Z_C}\left(1 - |\Gamma_L|^2\right)$$

(chap4.pdf, p.78)

This equals the power of the forward wave minus the power of the backward wave. *(chap4.pdf, p.78)*

1.6.2 Lossy Line --- Phasor Form *(chap4.pdf, pp.79--81)*

Equivalent circuit adds series resistance r [Ω/m] and shunt conductance g [S/m]:

Per-unit-length impedance and admittance:

$$z = r + j\omega l, \qquad y = g + j\omega c$$

(chap4.pdf, p.80)

The decoupled wave equation:

$$\frac{d^2\hat{V}}{dz^2} = \gamma^2\,\hat{V}, \qquad \gamma = \sqrt{zy} = \alpha + j\beta$$

(chap4.pdf, p.80)

Solution:

$$\hat{V}(z) = V^+\,e^{-\gamma z} + V^-\,e^{+\gamma z}$$

(chap4.pdf, p.81)

Characteristic impedance of lossy line:

$$Z_C = \sqrt{\frac{z}{y}} = \sqrt{\frac{r + j\omega l}{g + j\omega c}}$$

(chap4.pdf, p.81)

Both forward and backward traveling waves are **attenuated** exponentially as they travel. *(chap4.pdf, p.81)*

Note: $r(f)$ is frequency-dependent due to skin effect. *(chap4.pdf, p.80)*

1.7 4.7 Effect of Losses on Signal Integrity

1.7.1 Conductor Losses --- Circular Wire *(chap4.pdf, pp.82--83)*

DC resistance per unit length (current uniformly distributed over cross-section):

$$r_{dc} = \frac{1}{\sigma\,\pi r_w^2} \quad [\Omega/\text{m}]$$

(chap4.pdf, p.82)

Skin depth (current confined to outer annulus at high frequency):

$$\delta = \frac{1}{\sqrt{\pi f\,\mu\,\sigma}} \quad [\text{m}]$$

(chap4.pdf, p.82)

Skin depth **decreases** with the square root of frequency. *(chap4.pdf, p.82)*

High-frequency resistance per unit length (for $r_w \gg \delta$):

$$r_{hf} = \frac{1}{2\pi r_w\,\sigma\,\delta} = \frac{1}{2\pi r_w}\sqrt{\frac{\pi f\,\mu}{\sigma}} \quad [\Omega/\text{m}]$$

(chap4.pdf, p.83)

r_{hf} **increases** with $\sqrt{f}$ (10 dB/decade). *(chap4.pdf, p.83)*

1.7.2 Conductor Losses --- Rectangular PCB Land *(chap4.pdf, p.84)*

DC resistance per unit length (width w, thickness t):

$$r_{dc} = \frac{1}{\sigma\,w\,t} \quad [\Omega/\text{m}]$$

(chap4.pdf, p.84)

High-frequency resistance per unit length (current concentrated on the wide faces):

$$r_{hf} = \frac{1}{2\,\sigma\,w\,\delta} \quad [\Omega/\text{m}]$$

(chap4.pdf, p.84)

The turning frequency between r_{dc} and r_{hf} is obtained by setting $r_{dc} = r_{hf}$. In practice, current actually peaks at corners. *(chap4.pdf, p.84)*

1.7.3 Dielectric Losses *(chap4.pdf, pp.85--87)*

Dielectrics have bound charge (dipoles) but no appreciable free charge. Dielectric loss arises because dipoles cannot completely follow the time-varying electric field. *(chap4.pdf, p.85)*

Complex permittivity:

$$\varepsilon = \varepsilon' - j\varepsilon''$$

(chap4.pdf, p.86)

The capacitive per-unit-length admittance with complex ε:

$$y_c = j\omega\, c\, \frac{\varepsilon' - j\varepsilon''}{\varepsilon'} = \omega c\, \frac{\varepsilon''}{\varepsilon'} + j\omega c$$

(chap4.pdf, p.86)

Real part → per-unit-length **conductance**: $g = \omega c \tan\theta$. *(chap4.pdf, p.87)*

Loss tangent:

$$\tan\theta = \frac{\varepsilon''}{\varepsilon'}$$

(chap4.pdf, p.87)

Key: $g = \omega c \tan\theta$ → **conductance increases directly with frequency** (20 dB/decade), unlike skin-effect resistance (10 dB/decade). *(chap4.pdf, p.87)*

For FR-4 PCB material: $\tan\theta \approx 0.02$. *(chap4.pdf, p.94)*

1.7.4 Dispersion Effects *(chap4.pdf, pp.88--89)*

Two types of pulse distortion from line losses: - **Due to $\alpha(f)$ dispersion:** different frequency components are attenuated by different amounts → shape distortion. - **Due to $\beta(f)$ dispersion:** different frequency components travel at different speeds → arrival-time distortion.

Both reduce bandwidth: high-frequency components are attenuated/delayed more, increasing pulse rise/fall times. *(chap4.pdf, p.88)*

Power loss for a matched load over length L:

$$P_L = e^{-2\alpha L}$$

(chap4.pdf, p.89)

Loss is negligible when $r \ll \omega l$ and $g \ll \omega c$. *(chap4.pdf, p.89)*

1.7.5 Low-Loss Approximation *(chap4.pdf, pp.90--94)*

Criterion:

$$r \ll \omega l \quad \text{and} \quad g \ll \omega c$$

(chap4.pdf, p.90)

Under this condition, the propagation constant simplifies to: *(chap4.pdf, p.90)*

$$\alpha \approx \frac{r}{2Z_C} + \frac{gZ_C}{2} \quad [\text{Np/m}]$$

(chap4.pdf, p.91)

$$\beta \approx \omega\sqrt{lc} \quad [\text{rad/m}]$$

(chap4.pdf, p.91)

Characteristic impedance (frequency-independent):

$$Z_C \approx \sqrt{\frac{l}{c}} \quad [\Omega]$$

(chap4.pdf, p.91)

Sub-regions for typical PCB (FR-4): *(chap4.pdf, pp.92--94)*

Frequency	α behavior	β behavior
$<\sim 5$ MHz	Dominated by r_{dc} (constant)	Dispersive
5 MHz – -20 MHz	No significant dispersion	Linear with f
> 20 MHz	Skin-effect $\alpha \propto \sqrt{f}$	Linear with f (no dispersion)

Above 5 MHz: low-loss approximation is valid. Above 20 MHz: no β dispersion, but α dispersion (from skin effect) remains. *(chap4.pdf, p.93)*

Because $g = \omega c \tan\theta$ is always smaller than ωc by a factor of $\tan\theta \approx 0.02$, dielectric loss is dominated by conductor loss at lower frequencies but becomes comparable at very high frequencies. *(chap4.pdf, p.94)*

1.8 4.8 Lumped-Circuit Approximate Models *(chap4.pdf, pp.95--97)*

1.8.1 Criterion for Use *(chap4.pdf, p.95)*

The transmission line can be modeled as lumped elements only when the line is **electrically short** at the highest frequency of interest:

$$L < \frac{\lambda}{10}$$

(chap4.pdf, p.95)

1.8.2 Lumped-π and Lumped-T Models *(chap4.pdf, p.95)*

Two common approximations: - **Lumped-π**: half-capacitor --- inductor --- half-capacitor - **Lumped-T**: half-inductor --- capacitor --- half-inductor

Lossy lines are **difficult** to handle in time-domain with lumped models because the per-unit-length resistance is frequency-dependent. *(chap4.pdf, p.95)*

1.8.3 Accuracy vs. Number of Sections *(chap4.pdf, pp.96--97)*

Model	Valid up to
Distributed (exact)	All frequencies
Lumped-π, 1 section	f where $L = \lambda/10$
Lumped-π, 2 sections	Approximately $2\times$ the 1-section limit

Each additional section extends the valid frequency range by approximately the same factor. *(chap4.pdf, p.97)*